

Vertex Precision Engineering Ltd: Comprehensive Knowledge Base and Operations Manual

1. Company Overview

Company Name: Vertex Precision Engineering Ltd

Location: Unit 14, Black Country Enterprise Park, Dudley Road, Wolverhampton, West Midlands, WV2 4XD, United Kingdom

Year Founded: 2009

Total Employees: 18

Core Certifications: ISO 9001:2015, AS9100D

Company Description Vertex Precision Engineering Ltd is a premier manufacturer of high-precision CNC machined components, complex mechanical assemblies, and specialized engineering parts. Situated in the heart of the West Midlands—the traditional and modern hub of United Kingdom manufacturing—our facility combines decades of metallurgical expertise with state-of-the-art multi-axis CNC technology. Founded in 2009 by a team of former aerospace manufacturing engineers, Vertex was established to bridge the gap between high-volume, low-precision production and the prohibitive costs of ultra-high-precision bespoke machining. Today, we operate with a highly skilled workforce of 18 dedicated professionals, including CAD/CAM programmers, senior machinists, metrology specialists, and quality assurance personnel.

Mission Statement To deliver uncompromising precision and reliability in mechanical components, empowering the world's most demanding industries to innovate safely and efficiently. We achieve this through relentless adherence to quality standards, continuous investment in advanced machining technology, and a commitment to transparent, collaborative customer relationships.

Specialisms Vertex Precision Engineering specializes in the machining of complex geometries and tight-tolerance components from both standard and exotic materials. Our core specialisms include sub-contract 5-axis milling, precision CNC turning of hardened alloys, the manufacturing of custom gear profiles and rotors, and the production of mission-critical aerospace fasteners. We excel in low-to-medium volume production runs where quality and material traceability are paramount.

Industries Served We proudly serve industries where failure is not an option. Our primary markets include:

- **Aerospace:** Structural components, landing gear bushings, engine mounts, and specialized fasteners.

- **Automotive (High-Performance & Motorsport):** Gearbox components, hydraulic manifolds, and custom bearings.
- **Oil & Gas:** Subsea valve components, high-pressure fittings, and corrosion-resistant shafts.
- **Medical Devices:** Surgical instrumentation, implantable grade titanium components, and pump rotors.
- **Defence:** Ruggedized housings, weapon system components, and secure communications hardware.
- **Renewable Energy:** Wind turbine gearbox components, solar panel actuation shafts.

2. Product Catalogue

Our product catalogue encompasses a wide range of precision-engineered components. We supply both standard catalogued dimensions and bespoke items manufactured to exact customer drawings.

2.1 CNC Turned Parts

CNC turning is the foundation of our rotational component manufacturing. Using advanced multi-axis lathes with live tooling, we produce cylindrical parts with exceptional concentricity and surface finish.

- **Shafts:** Transmission shafts, drive shafts, and armature shafts for motors. We ensure precise runout and dynamic balance.
- **Pins:** Dowel pins, shear pins, and clevis pins used for critical alignment and load-bearing applications.
- **Bushings:** Plain bearings and isolation bushings designed for high-wear environments.
- **Studs:** Fully threaded, double-ended, and custom-collared studs for high-stress fastening.

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| **Specification | Details | | Materials Available** | Steels (EN8, EN24, EN36, 304, 316, 17-4PH), Aluminium (6082, 7075, 2024), Titanium (Gr 2, Gr 5), Exotics (Inconel, Hastelloy), Plastics (PEEK, Nylon, Delrin) | | **Tolerances** | $\pm 0.005\text{mm}$ (Standard), $\pm 0.002\text{mm}$ (Precision) | | **Lead Time** | 10–15 Working Days | | **Minimum Order Qty** | 10 units | | **Price Range (GBP)** | £4.50 – £250.00 per unit (geometry/material dependent) | | **Common Applications** | Motor armatures, gearbox internals, structural alignments, hydraulic cylinders. |

2.2 CNC Milled Parts

Our 3-axis, 4-axis, and 5-axis milling centres allow us to produce prismatic parts with complex 3D profiles and deep cavities in a single setup, minimizing handling errors.

- **Brackets:** Structural mounting brackets optimized for weight reduction and high stiffness.
- **Plates:** Base plates, fixture plates, and manifold plates requiring high flatness and parallelism.
- **Housings:** Electronic enclosures, gearbox casings, and sensor housings requiring EMI shielding and environmental sealing.
- **Flanges:** Mounting flanges and pipe flanges, including ANSI and custom bolt circles.

| **Specification** | **Details** | | **Materials Available** | All standard Steels, Aluminium alloys, Titanium, Exotic alloys, and Plastics. | | **Tolerances** | $\pm 0.005\text{mm}$ (Standard), $\pm 0.002\text{mm}$ (Precision) | | **Lead Time** | 12–18 Working Days | | **Minimum Order Qty** | 5 units | | **Price Range (GBP)** | £25.00 – £850.00 per unit | | **Common Applications** | Aerospace structural nodes, motorsport subframes, radar array mounts. |

2.3 Precision Bearings

We manufacture specialty and low-volume precision bearings where off-the-shelf solutions fail due to extreme temperatures, corrosive environments, or non-standard load ratings. Sizes range from 6mm to 200mm bore.

- **Ball Bearings:** Deep groove and angular contact ball bearings for high-speed applications.
- **Roller Bearings:** Cylindrical and tapered roller bearings for heavy radial loads.
- **Thrust Bearings:** Designed specifically to handle high axial loads in rotary systems.

| **Specification** | **Details** | | **Materials Available** | High-carbon steels (EN36), 440C Stainless, PEEK (for cages) | | **Tolerances** | ABEC 5 to ABEC 9 equivalents ($\pm 0.002\text{mm}$) | | **Lead Time** | 20–25 Working Days | | **Minimum Order Qty** | 20 units | | **Price Range (GBP)** | £35.00 – £1,200.00 per unit | | **Common Applications** | Jet engine main shafts, high-performance machine tool spindles, subsea pumps. |

2.4 Rotors

Rotor manufacturing requires specialized CAM programming to achieve complex blade geometries and ensure perfect aerodynamic or hydrodynamic balance.

- **Turbine Rotors:** High-temperature resistant rotors for gas turbines and micro-turbines.
- **Pump Rotors:** Helical and lobe rotors for positive displacement and centrifugal pumps.
- **Compressor Rotors:** Impellers and blisks (blade integrated disks) for air and gas compression.

| **Specification** | **Details** | | **Materials Available** | Titanium Grade 5, Inconel 625, Inconel 718, 17-4PH Stainless | | **Tolerances** | $\pm 0.005\text{mm}$, with rigorous dynamic balancing | | **Lead Time** | 25–35 Working Days | | **Minimum Order Qty** | 1 unit | | **Price Range (GBP)** | £450.00 – £5,500.00 per unit | | **Common Applications** | Jet engines, industrial HVAC chillers, heavy-duty fluid transfer systems. |

2.5 Gears

We offer comprehensive gear cutting, hobbing, and shaping services for power transmission applications. Available from module 0.5 to module 8.

- **Spur Gears:** Standard straight-cut gears for efficient, parallel shaft power transfer.
- **Helical Gears:** Angled teeth for smoother, quieter operation under heavy loads.
- **Bevel Gears:** Straight and spiral bevels for changing the axis of rotational power (typically 90 degrees).
- **Worm Gears:** High-reduction ratio gear sets for self-locking mechanisms.

| **Specification** | **Details** | | **Materials Available** | EN24, EN36 (Case Hardened), Bronze (for worm wheels), Delrin (low load) | | **Tolerances** | DIN Quality Class 5 to 7 | | **Lead Time** | 15–25 Working Days | | **Minimum Order Qty** | 5 units | | **Price Range (GBP)** | £40.00 – £900.00 per unit | | **Common Applications** | Automotive transmissions, robotic articulators, heavy winch systems. |

2.6 Hydraulic Fittings and Manifolds

Fluid power components require immaculate internal surface finishes to prevent turbulent flow and high-pressure blowouts.

- **Fittings:** Custom adapters, bulkheads, and high-pressure JIC/UNF/Metric fittings.
- **Manifolds:** Complex, multi-port solid blocks designed to regulate fluid flow between pumps and actuators.

| **Specification** | **Details** | | **Materials Available** | 316 Stainless Steel, Aluminium 6082, Hastelloy C276 | | **Tolerances** | $\pm 0.005\text{mm}$, NPT/BSP thread precision | | **Lead Time** | 10–

20 Working Days | | **Minimum Order Qty** | 10 units (Fittings), 1 unit (Manifolds) | | **Price Range (GBP)** | £15.00 – £1,500.00 per unit | | **Common Applications** | Deep-water ROVs, agricultural machinery, aerospace control surfaces. |

2.7 Fasteners (Aerospace-Grade)

We produce specialized fastening hardware that exceeds standard commercial grades, accompanied by full FAIR and material certification.

- **Bolts:** Hex, 12-point, and socket head cap screws engineered for extreme tensile strength.
- **Nuts:** Self-locking, castle, and flange nuts.
- **Washers:** Load-spreading, Belleville, and custom sealing washers.

| **Specification** | **Details** | | **Materials Available** | Titanium Grade 5, Inconel 718, 17-4PH, 316 Stainless | | **Tolerances** | Class 3A/3B thread fit, ± 0.005 mm dimensional | | **Lead Time** | 10–15 Working Days | | **Minimum Order Qty** | 100 units | | **Price Range (GBP)** | £5.00 – £85.00 per unit | | **Common Applications** | Spacecraft assemblies, high-temperature exhaust mounting, critical structural joints. |

2.8 Custom Fabricated Assemblies

Beyond individual components, Vertex offers complete electro-mechanical and purely mechanical assembly services. This includes pressing bearings, installing Helicoils/inserts, welding, and final tolerance checking of the mated assembly.

| **Specification** | **Details** | | **Materials Available** | Mixed material assemblies per drawing | | **Tolerances** | Stack-up tolerance analysis provided, ± 0.005 mm | | **Lead Time** | 20–40 Working Days | | **Minimum Order Qty** | 1 assembly | | **Price Range (GBP)** | £200.00 – £15,000.00+ (quoted per project) | | **Common Applications** | Turnkey sub-assemblies for medical scanners, automated manufacturing jigs. |

3. Materials & Specifications

Vertex Precision Engineering works with a strictly controlled supply chain to source fully certified raw materials. All metals and plastics are stored in climate-controlled conditions and catalogued for 100% traceability.

3.1 Steels

Steel remains the backbone of mechanical engineering due to its versatility, cost-effectiveness, and heat-treatability.

- **EN8 (080M40):** Unalloyed medium carbon steel. Good tensile strength. Typical Use: Axles, general shafts, stressed pins.

- **EN24 (817M40):** High tensile alloy steel. Typical Use: High-strength gears, heavy-duty drive shafts.
- **EN36 (655M13):** Carburizing steel. Provides a very hard outer case with a tough core. Typical Use: Heavy-duty gears, thrust washers.
- **304 Stainless Steel:** The standard austenitic stainless steel. Excellent corrosion resistance. Typical Use: Food processing parts, standard brackets.
- **316 Stainless Steel:** Molybdenum added for superior pitting and marine corrosion resistance. Typical Use: Oil & gas fittings, marine components.
- **17-4PH Stainless:** Precipitation hardened stainless steel combining high strength and corrosion resistance. Typical Use: Aerospace structural parts, high-stress medical tools.

3.2 Aluminium Alloys

Preferred for applications requiring high strength-to-weight ratios.

- **6082:** A medium strength structural alloy with excellent corrosion resistance. Typical Use: Highly stressed applications, trusses, bridges, cranes.
- **7075:** Zinc is the primary alloying element. Extremely high strength, comparable to many steels. Typical Use: Aerospace wing spars, motorsport bulkheads.
- **2024:** Copper alloyed. High yield strength and good fatigue resistance. Typical Use: Aircraft structures, clock parts, computer parts.

3.3 Titanium

Exceptional strength, low weight, and biocompatibility.

- **Grade 2:** Commercially pure titanium. Excellent corrosion resistance and formability. Typical Use: Chemical processing, medical implants.
- **Grade 5 (Ti-6Al-4V):** The workhorse titanium alloy. High strength and heat resistance. Typical Use: Jet engine compressor blades, aerospace fasteners.

3.4 Exotic Alloys

Designed to survive extreme temperatures, aggressive chemicals, and immense stress.

- **Inconel 625:** Nickel-chromium-molybdenum alloy. Outstanding fatigue strength and resistance to chloride-ion stress-corrosion cracking. Typical Use: Marine exhausts, subsea wellhead components.
- **Inconel 718:** High-strength, corrosion-resistant nickel chromium material used at -423° to 1300°F. Typical Use: Gas turbine engine parts, cryogenic tanks.

- **Hastelloy C276:** Tungsten-added nickel-molybdenum-chromium alloy. Ultimate resistance to severe corrosion. Typical Use: Chemical reactors, sour gas extraction valves.

3.5 Engineering Plastics

Used where electrical insulation, low friction, or chemical resistance is required without the weight of metal.

- **PEEK:** Organic thermoplastic polymer. Incredible heat and chemical resistance. Typical Use: Medical implants, high-temp aerospace insulators.
- **Nylon PA66:** High mechanical strength, rigidity, and good stability under heat. Typical Use: Gears, bearings, wear pads.
- **Delrin (POM-C):** Acetal resin with low friction and high wear resistance. Typical Use: Precision gears, electrical insulator components, snap-fit assemblies.

Material Specifications Summary Table

Material Category	Grade	Machinability Rating	Available Stock Sizes (Bar/Billet)
Typical Use Case	Steel	EN8 65%	5mm - 300mm General shafts, structural pins
Steel	EN24	50%	10mm - 250mm High-strength gears
Steel	EN36	45%	10mm - 200mm Case-hardened components
Steel (SS)	304 SS	45%	3mm - 400mm Food grade components
Steel (SS)	316 SS	40%	3mm - 400mm Marine fittings
Steel (SS)	17-4PH	35%	10mm - 150mm Aerospace structural joints
Aluminium	6082	90%	5mm - 500mm Standard brackets, manifolds
Aluminium	7075	80%	10mm - 300mm High-stress aero plates
Aluminium	2024	85%	10mm - 250mm Fatigue-resistant components
Titanium	Grade 2	40%	5mm - 150mm Medical implants
Titanium	Grade 5	30%	5mm - 200mm Turbine blades, aero fasteners
Exotic	Inconel 625	15%	10mm - 100mm Extreme heat exhausts
Exotic	Inconel 718	12%	10mm - 100mm Gas turbine components
Exotic	Hastelloy C276	15%	10mm - 80mm Corrosive chemical valves
Plastic	PEEK	60%	5mm - 100mm High-temp insulators
Plastic	Nylon PA66	80%	5mm - 200mm Wear pads, low-load gears
Plastic	Delrin	95%	5mm - 250mm Precision plastic gears

4. Manufacturing Capabilities

Our West Midlands facility houses advanced, rigorously calibrated manufacturing equipment to support both subtractive manufacturing and precision verification.

4.1 CNC Turning

Our turning department utilizes multi-axis CNC lathes capable of continuous production as well as large, single-item prototyping. Live tooling capabilities allow us to mill, drill, and tap cross-holes without moving the part to a separate milling machine.

- **Maximum Turning Diameter:** 500mm
- **Maximum Turning Length:** 2000mm
- **Bar Feeder Capacity:** Up to 65mm through-spindle.
- **Tolerances:** ± 0.005 mm on diameters.

4.2 CNC Milling

We utilize vertical and horizontal machining centres, including full simultaneous 5-axis capability. This allows for complex geometries, undercuts, and off-angle boring in a single setup.

- **5-Axis Work Envelope:** 800mm (X) x 600mm (Y) x 500mm (Z)
- **3-Axis Work Envelope:** 1200mm (X) x 800mm (Y) x 600mm (Z)
- **Spindle Speeds:** Up to 18,000 RPM (ideal for high-speed aluminium and plastic machining).
- **Tolerances:** ± 0.005 mm positional accuracy.

4.3 Grinding

When tolerances tighter than ± 0.005 mm are required, or when working with hardened steels, our grinding department ensures perfect dimensional and surface finishing.

- **Surface Grinding:** Flatness and parallelism to within 0.002mm. Magnetic chuck capacity of 600mm x 300mm.
- **Cylindrical Grinding:** OD (Outside Diameter) and ID (Inside Diameter) grinding for shafts and bearings. Tolerances achievable: ± 0.002 mm.

4.4 EDM Wire Cutting

Electrical Discharge Machining (EDM) is utilized for cutting intricate internal shapes (like splines or keyways) and for machining ultra-hard conductive materials (like Inconel or hardened tool steel) without introducing mechanical stress.

- **Cutting Envelope:** 400mm x 300mm x 250mm
- **Wire Diameter:** 0.1mm to 0.25mm
- **Tolerances:** ± 0.002 mm.

4.5 Quality Assurance & Metrology

Our temperature-controlled inspection laboratory is the final gatekeeper for all Vertex products.

- **CMM Inspection:** Automated Coordinate Measuring Machines (CMM) utilizing Renishaw probing systems. Capable of scanning complex 3D profiles to verify against native CAD models.
- **Optical Comparators:** For profile and thread measurement.
- **Surface Finish:** Standard achievable finish is Ra 0.4 to Ra 3.2. We use calibrated profilometers to verify Ra, Rz, and Rq values.

5. Industries Served

Vertex Precision Engineering operates across sectors where mechanical failure results in catastrophic downtime or safety risks. We align our manufacturing processes with the strict compliance frameworks of these industries.

| Industry | Example Parts Supplied | Compliance / Quality Requirements Met |
Aerospace | Landing gear struts, structural ribs, Inconel exhaust nozzles, Grade 5 Titanium fasteners. | AS9100D, FAIR (AS9102), Full Material Traceability, ITAR/EAR compliance where necessary. | **Automotive (Motorsport)** | Differential housings, custom gear sets, suspension bellcranks, engine valves. | PPAP (Production Part Approval Process), ISO 9001:2015, Material Certification. | **Oil & Gas** | High-pressure directional drilling components, Hastelloy subsea valves, pump impellers. | NACE MR0175/ISO 15156 (sour gas environments), API standards compliance. | **Medical Devices** | Bone saws, implantable Titanium screws, centrifuge rotors, surgical instrument handles. | Machining aligned with ISO 13485 principles, biocompatible fluid protocols, ultrasonic cleaning. | **Defence** | Radar mounting brackets, secure communications encasements, heavy vehicle drive shafts. | UK MOD Defence Standards (Def Stan), strict NDA and secure facility protocols. | **Renewable Energy** | Wind turbine gearbox shafts, solar tracking actuator pins, hydroelectric pump impellers. | ISO 9001:2015, Environmental stress testing documentation, heavy load certs. |

6. Ordering Process

We have streamlined our ordering process to ensure rapid, accurate quoting and seamless transition from design to production.

Step 1: Request a Quote (RFQ) Send your RFQ via email to sales@vertexprecision.co.uk or use our secure online portal.

Step 2: Submit Files For accurate quoting and manufacturing, please supply:

- **3D CAD Models:** STEP (.step / .stp) or IGES format (STEP is highly preferred).
- **2D Technical Drawings:** PDF format. This is crucial for detailing critical tolerances, surface finish requirements, thread specifications, and material treatments that a 3D model cannot convey.

Step 3: Quote Generation & DFM Review Our engineering team will review your files. If we identify features that are difficult or expensive to machine, we will provide a Design for Manufacturability (DFM) report with suggestions to reduce costs. A formal quote is usually provided within 24–48 hours.

Step 4: Order Confirmation & Lead Times Upon accepting the quote and issuing a Purchase Order (PO):

- **Standard Lead Time:** 10–15 working days.
- **Express/Expedited Lead Time:** 5 working days (subject to machine availability and a premium surcharge).

Step 5: Manufacturing & Quality Control Parts are machined, inspected throughout the run, and undergo final CMM inspection.

Step 6: Payment Terms & Dispatch

- **New Customers:** Payment via Credit/Debit Card or Bank Transfer (BACS/CHAPS) upon pro-forma invoice prior to dispatch.
- **Approved Accounts:** 30 days net from the date of invoice.
- Parts are carefully packaged to prevent transit damage and shipped via next-day domestic or tracked international couriers.

7. Quality & Certifications

Quality is not just an inspection step at Vertex; it is integrated into every stage of our workflow.

- **ISO 9001:2015:** We are fully certified to the internationally recognized Quality Management System standard. This ensures consistent processes, continuous improvement, and a strong customer focus.
- **AS9100D:** This is the aerospace industry standard for quality assurance, building upon ISO 9001. It mandates stringent controls over supplier management, product traceability, counterfeit parts prevention, and risk management.
- **Material Traceability:** We provide full EN 10204 Type 3.1 material certificates for all raw materials upon request, tracing the exact chemical and mechanical properties back to the original steel or alloy mill.
- **First Article Inspection Report (FAIR):** For aerospace, defence, and critical components, we provide a comprehensive FAIR (formatted to AS9102 standards if required). This verifies that every single dimension on the 2D drawing has been independently checked and recorded on the first manufactured part before the full production run begins.

- **Inspection Reports:** Standard dimensional inspection reports are provided with all shipments, confirming parts meet the stated geometric dimensioning and tolerancing (GD&T) requirements.

8. FAQ Section

1. What is your standard Minimum Order Quantity (MOQ)? For standard CNC milled and turned parts, our MOQ is generally 5 to 10 units. However, for complex prototyping, custom rotors, and large bespoke fabricated assemblies, the MOQ is 1 unit. Aerospace fasteners require an MOQ of 100 units due to setup times.

2. How do I send drawings securely to Vertex? Drawings can be emailed directly to our sales team. For sensitive defence or proprietary designs, we offer a secure, encrypted FTP portal. We also accept files sent via secure links (e.g., WeTransfer) provided they are password protected.

3. What file formats do you accept for quoting and manufacturing? We require a 3D model, ideally in STEP (.stp) format, though IGES and Parasolid are acceptable. Crucially, we also need a 2D drawing in PDF format to understand specific tolerances, threaded holes, and surface finishes.

4. What are your standard lead times? Our standard production lead time is 10 to 15 working days from receipt of a Purchase Order and clean CAD files. This can vary slightly depending on material availability and the complexity of the part.

5. Do you offer an express or expedited service? Yes. We offer an express 5-working-day lead time for urgent orders. This service incurs an expedite fee (typically 30-50% on top of the standard quote) and is subject to current machine capacity and immediate material availability.

6. How is pricing calculated for a quote? Pricing is based on material costs, estimated machine cycle time, setup complexity (number of operations required), tooling wear (especially high for Exotics like Inconel), and any requested post-processing (e.g., anodizing, heat treatment).

7. Can you provide material selection advice? Absolutely. Our engineering team frequently assists clients in optimizing material choices. For example, if a client requests 7075 Aluminium but the part requires extensive welding, we will advise switching to 6082 Aluminium for better weldability.

8. What is the tightest tolerance Vertex can achieve? Our standard CNC machining tolerance is $\pm 0.005\text{mm}$. However, utilizing our precision cylindrical and surface grinding equipment, we can achieve tolerances down to $\pm 0.002\text{mm}$.

9. What surface finishes are possible? Straight off the CNC machine, we typically achieve a surface finish of Ra 1.6 to Ra 3.2. With finishing passes, we can achieve Ra 0.8. Using grinding or polishing, we can bring the finish down to Ra 0.4 or lower.

10. Do you ship internationally? Yes, we serve a global client base. International shipping is handled via trusted couriers (DHL, FedEx, UPS) utilizing Incoterms (usually EXW or DAP). Delivery times vary by destination and customs clearance.

11. What is your domestic shipping policy within the UK? For UK deliveries, we use fully insured next-working-day courier services. Parts are securely boxed, vacuum-sealed if prone to rust, and packed with custom foam inserts for delicate assemblies.

12. What happens if there is a quality dispute or I need to return parts? Under our ISO 9001 procedures, if parts do not meet the agreed-upon 2D drawing specifications, we will issue an RMA (Return Merchandise Authorization). We will investigate the root cause, and rework or remanufacture the non-conforming parts at our expense.

13. How do you handle repeat orders? All CNC programs, setup sheets, and tooling lists are securely archived under your specific Part Number and Revision ID. Repeat orders skip the CAM programming phase, often resulting in slightly faster lead times and perfectly consistent parts.

14. What is the difference between prototype and production runs? Prototype runs (1-4 parts) prioritize speed and testing, often without optimization of tool paths, making the per-unit cost higher. Production runs prioritize cycle-time efficiency, custom fixture creation, and economies of scale, drastically lowering the per-unit cost.

15. Will you sign a Non-Disclosure Agreement (NDA)? Yes. We routinely sign NDAs before clients share confidential CAD files or intellectual property. We have standard NDAs available, or we can review and sign your company's preferred legal document.

16. Are you compliant with Export Control laws (ITAR/EAR)? Yes. While UK-based, we are experienced in handling Export Administration Regulations (EAR) and International Traffic in Arms Regulations (ITAR) requirements for our US defence and aerospace clients, ensuring secure data handling and restricted facility access.

17. Can you source specialty materials not listed in your catalogue? Yes. While our catalogue lists our most common materials, our purchasing department has strong relationships with global alloy suppliers. We can source specific grades of maraging steel, specific titanium composites, or rare plastics upon request.

18. Do you offer heat treatment services? We do not perform heat treatment in-house. However, we project-manage the entire process through our trusted, Nadcap-approved sub-contractors in the West Midlands. We handle the logistics and supply the final certified, hardened part.

19. Do you offer surface treatments like anodizing or plating? Similar to heat treatment, we manage surface finishing (Hard Anodizing, Zinc Plating, Electroless Nickel Plating, Passivation, etc.) through fully vetted, ISO-certified local partners, delivering a finished, ready-to-use component.

20. What is a First Article Inspection Report (FAIR)? A FAIR is a comprehensive document verifying that a production process yields parts meeting all engineering requirements. We measure every dimension, note, and specification on the drawing for the first part produced and record it, usually to AS9102 format for aerospace.

21. Do I need a FAIR for every order? No. FAIRs are generally only required for the first production run of a new aerospace/defence part, or if there has been a revision to the drawing, a change in manufacturing location, or a lapse in production of over two years.

22. How do I apply for a 30-day net credit account? After your first three successfully completed and paid-upfront orders, you can request a credit application form. Our accounts team will run standard commercial credit checks before approving a 30-day net facility.

23. What currency are your quotes in? By default, all quotes are in Great British Pounds (GBP / £). We can provide quotes in USD (\$) or EUR (€) upon specific request, valid for 14 days due to exchange rate fluctuations.

24. Can you reverse engineer an existing broken part? Yes. We can use our CMM equipment and manual metrology tools to measure a physical sample, create a new 3D CAD model, generate manufacturing drawings, and produce a replacement part.

25. Do you manufacture gears in-house? Yes, we have dedicated gear hobbing, shaping, and cutting machinery in-house, capable of producing spur, helical, bevel, and worm gears up to module 8.

26. What happens if I revise my design after I place an order? If the order is already in production, design changes may incur scrap costs for ruined material and machine time. If caught before production starts, we will review the new CAD, update the quote if the complexity has changed, and issue a revised timeline.

27. Do you stamp or laser engrave part numbers? Yes. We offer dot-peen marking, laser etching, and CNC engraving. We can add part numbers, revision letters, batch codes, or company logos exactly where specified on your 2D drawing.

28. How long is your quotation valid for? Due to fluctuating raw material costs (especially for titanium and exotic alloys), our standard quotations are valid for 30 days.

29. Can I visit your facility in the West Midlands? We welcome existing and prospective clients to audit our facility. Due to confidentiality and safety regulations, visits must be scheduled in advance with our management team.

30. What is your maximum machining capacity for turning? Our largest lathes can handle billets up to 500mm in diameter and shaft lengths up to 2000mm.

31. What is your maximum machining capacity for milling? Our largest 3-axis mill has an envelope of 1200mm x 800mm x 600mm. Our 5-axis machines can accommodate parts up to 800mm x 600mm x 500mm.

32. Do you charge separately for tooling and fixtures? Standard cutting tools are absorbed into the hourly machining rate. However, if your part requires custom work-holding fixtures, highly specialized threading taps, or custom form tools, these will be itemized as a one-off NRE (Non-Recurring Engineering) cost on your quote.

33. What is NRE? Non-Recurring Engineering (NRE) refers to the one-time costs to research, design, develop, and test a new product or part. At Vertex, this usually covers CAM programming time and custom fixture manufacturing. You only pay this on the first order.

34. Can you handle blanket orders with scheduled call-offs? Yes. We frequently set up scheduled deliveries for our OEM clients. You can place a blanket order for 1,000 units, and we will manufacture the batch and hold them in our inventory, releasing 100 units a month over 10 months.

35. How do you prevent counterfeit materials? In compliance with AS9100D, we only purchase raw materials from approved, certified distributors. Every batch of material entering our facility must be accompanied by original mill certificates, which are verified against our purchase orders.

36. Do you do structural welding? We offer TIG and MIG welding for custom fabricated assemblies. Our welders are coded to BS EN ISO 9606 standards for steel, stainless steel, and aluminium.

37. What happens if my part requires assembly with components you don't manufacture? We provide full mechanical assembly services. If your BOM (Bill of Materials) includes commercial off-the-shelf (COTS) items like specific springs, circlips, or electronic sensors, we can procure those and integrate them into the final assembly.

38. How do you manage tool wear to maintain tolerances during a long production run? Our CNC machines use automated tool pre-setters and touch-probes. The machine routinely checks the cutting tool for wear or breakage and automatically offsets the cutting path to compensate for micro-wear, ensuring the 1000th part is exactly the same as the 1st.

39. Are your inspection instruments calibrated? Yes. All metrology equipment, from our CMMs down to individual micrometers and thread gauges, is calibrated annually to national standards (UKAS), with a fully documented calibration matrix maintained by our Quality Manager.

40. Who is my primary point of contact once an order is placed? You will be assigned a dedicated Project Engineer. They will be your single point of contact from initial DFM review through to final delivery, ensuring clear communication and technical continuity.

9. Contact & Location

Vertex Precision Engineering Ltd is strategically located in the West Midlands, providing excellent transport links to the rest of the UK and major export hubs.

Headquarters and Manufacturing Facility: Vertex Precision Engineering Ltd Unit 14, Black Country Enterprise Park Dudley Road Wolverhampton, West Midlands WV2 4XD United Kingdom

Contact Information:

- **Main Switchboard / Sales:** +44 (0)1902 555 721
- **Sales & RFQ Email:** sales@vertexprecision.co.uk
- **Quality & Engineering Support:** engineering@vertexprecision.co.uk
- **Accounts & Billing:** accounts@vertexprecision.co.uk

Operating Hours:

- **Monday to Thursday:** 07:30 - 17:00
- **Friday:** 07:30 - 13:00
- **Saturday & Sunday:** Closed

Emergency AOG / Urgent Orders Contact: For Aerospace AOG (Aircraft on Ground) or critical production-line-down emergencies occurring outside standard operating hours, please contact our emergency production manager:

- **24/7 Emergency Line:** +44 (0)7700 900 832
- *Please note: Use of the emergency production protocol incurs immediate premium surcharges.*

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